

Specific stimulation of Graves' disease thyrocytes by the natural anti-Gal antibody from normal and autologous serum.

Anti-Gal is a natural Ab abundantly produced in humans. It interacts specifically with the carbohydrate epitope Gal alpha 1-3Gal beta 1-4GlcNAc-R (termed the alpha-galactosyl epitope). This epitope is expressed in large amounts on thyrocytes of nonprimate mammals, but not of humans. We have previously found that binding of anti-Gal to alpha-galactosyl epitopes on porcine thyrocytes results in stimulatory effects similar to those exerted by thyroid-stimulating hormone (thyrotropin). In the present study, we tested the hypothesis that anti-Gal may contribute to Graves' disease (GD) pathogenesis by stimulation of the thyrocytes of patients with this autoimmune disorder. Anti-Gal binding and stimulatory effects were assessed in primary thyrocyte cultures. Anti-Gal specifically bound to GD thyrocytes and induced an increase in cAMP synthesis, ^{125}I uptake, and DNA synthesis in these cells. Furthermore, the stimulatory effects of autologous sera on GD thyrocytes were greatly reduced after specific depletion of anti-Gal from these sera. No binding and no stimulatory effects of anti-Gal were observed, however, with normal human thyrocytes and with thyrocytes from thyrotoxic patients who lack thyroid-stimulating Igs or thyrotropin binding inhibiting Igs. These in vitro stimulatory effects of anti-Gal on GD thyrocytes suggest that this natural Ab may contribute to the in vivo continuous stimulation of thyrocytes in GD patients. The possibility that anti-Gal may stimulate GD thyrocytes via interaction with aberrantly expressed alpha-galactosyl epitopes on the thyroid-stimulating hormone receptor is discussed.